MULTIVARIATE ANALYSIS OF THE WINE QUALITY DATASET

An Integrated Study of Red and White Portuguese Vinho Verde Wines

APM1220 – Applied Multivariate Data Analysis

Far Eastern University
Summative Assessment 1

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1. Introduction

Wine characteristics arise from a combination of chemical composition and production processes. Physicochemical measurements—including acidity, sugar, chloride concentration, sulfur dioxide, density, pH, sulphates, and alcohol—provide objective descriptions of wine composition. These measurements are relevant to fermentation, preservation, and the chemical profile of a finished wine. Sensory quality, however, is evaluated by people and cannot be fully represented by chemical measurements alone.

This study examines the UCI Wine Quality Dataset for Portuguese Vinho Verde wines, combining red and white wine observations. It applies three multivariate methods with complementary purposes: Multivariate Analysis of Variance (MANOVA) tests whether red and white wines differ across selected chemical measurements; Principal Component Analysis (PCA) reduces the 11 physicochemical variables to a smaller set of dimensions; and Exploratory Factor Analysis (EFA) explores latent dimensions that may account for shared correlations among the measurements.

The objective is to identify the major physicochemical differences between wine colors, summarize the dominant patterns of variation, and assess whether those patterns can be interpreted as broader chemical constructs. Because the data are observational, the findings are interpreted as associations and descriptive structure rather than causal evidence.

2. Dataset

The study uses the Wine Quality Dataset made available through the UCI Machine Learning Repository. The source provides separate datasets for red and white Portuguese Vinho Verde wines. The two datasets were combined into one analytical table and a categorical color variable was retained to identify each wine as Red or White.

Dataset characteristic	**Description**
Total observations	6,497 wine samples
Red wine samples	1,599
White wine samples	4,898
Physicochemical variables	11 continuous measurements
Additional variables	quality score and wine color
Missing values	None detected in the combined dataset
Exact duplicate rows	1,177 (18.1%) in the full dataset; duplicates retained in the primary analysis

The 11 physicochemical variables are fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. The dataset also includes a sensory quality score and a categorical wine-color variable. The quality score is not used as an active variable in the MANOVA, PCA, or EFA analyses reported here.

The notebook retains exact duplicate rows in the primary analysis because no wine identifier is available and identical rounded laboratory measurements do not necessarily prove that two rows represent the same physical wine. A duplicate-free dataset is used as a sensitivity check for MANOVA. The 151 zero values in citric acid are treated as plausible measurements rather than automatically classified as missing-value placeholders.

3. Research Questions and Hypotheses

3.1 MANOVA: Group differences

Research question 1: Do red and white wines differ significantly in their combined physicochemical characteristics?

H_0: The population mean vectors for red and white wines are equal across the six selected variables: fixed acidity, volatile acidity, residual sugar, chlorides, sulphates, and total sulfur dioxide (μRed = μWhite).

H_1: The population mean vectors differ (μRed ≠ μWhite); at least one linear combination of the six variables differs between the groups.

3.2 PCA: Dimension reduction

Research question 2: Can the 11 physicochemical measurements be represented by a smaller number of principal components while retaining a substantial proportion of total variance?

PCA is primarily a descriptive dimension-reduction method, so it does not use a conventional null-versus-alternative hypothesis test for the number of components. The analytical decision is to retain components supported by the eigenvalue criterion, scree plot, and parallel analysis, then report their cumulative explained variance.

3.3 EFA: Latent dimensions

Research question 3: What latent dimensions may explain the observed correlations among the 11 physicochemical characteristics?

EFA is exploratory. Its working premise is that correlations among the observed variables may be represented by a smaller number of common factors plus variable-specific variance and error. Factorability is evaluated using the correlation structure, Bartlett's test of sphericity, and the Kaiser–Meyer–Olkin (KMO) measure.

4. Methodology

4.1 Data preparation and exploratory analysis

The red and white datasets were imported with pandas, combined into a single dataframe, and assigned a categorical color label. The analysis inspected dataset dimensions, variable types, missingness, exact duplicates, descriptive statistics, coefficients of variation, skewness, kurtosis, correlations, and potential univariate and multivariate outliers. Boxplots, density plots, correlation heatmaps, and within-color correlation matrices were used to visualize distributions and group differences.

Mahalanobis distances were used to flag potential multivariate outliers. These observations were not silently deleted; sensitivity analyses were used to examine whether the main conclusions changed when flagged observations were excluded.

4.2 MANOVA

MANOVA tested the equality of the red and white group mean vectors for six selected dependent variables. Variables were selected to represent different chemical domains and to avoid excessive redundancy: fixed acidity, volatile acidity, residual sugar, chlorides, sulphates, and total sulfur dioxide. Free sulfur dioxide was excluded because it correlated with total sulfur dioxide; density was excluded because of high variance inflation and overlap with alcohol and sugar; alcohol was excluded because its group difference was small.

The analysis reported Wilks' Lambda and Pillai's Trace, followed by univariate ANOVAs and Welch tests. Bonferroni adjustment was applied to the six follow-up tests ($\alpha = .05/6 \approx .0083$). Assumptions were evaluated using group-specific multivariate normality diagnostics, Mahalanobis outliers, Box's M, Levene tests, and within-group correlations. Robustness checks included log-transformed dependent variables, duplicate removal, outlier removal, and a permutation test.

4.3 Principal Component Analysis

PCA used all 11 physicochemical variables, excluding color and quality as active variables. Because the measurements have different units and scales, variables were standardized to z-scores before PCA; this is equivalent to performing PCA on the correlation matrix. The number of components was evaluated using eigenvalues greater than one, the scree plot, and parallel analysis. Component loadings were used to interpret the retained axes, and a Varimax rotation was also examined for interpretability. Robustness checks considered log-transformed variables and the removal of the red–white mean difference.

4.4 Exploratory Factor Analysis

EFA used the same 11 standardized physicochemical variables. Factorability was evaluated using Bartlett's test and KMO. Three factors were retained based on the Kaiser criterion, standard parallel analysis, the scree plot, fit, and interpretability. Principal Axis Factoring (PAF) was used because the variables showed skewness and heavy tails, making a normal-theory maximum-likelihood approach less suitable. Promax oblique rotation was chosen because chemical dimensions may be correlated. The analysis inspected pattern loadings, communalities, factor correlations, and potential cross-loadings, and compared solutions under log transformation, removal of density, removal of the color effect, and a four-factor alternative.

4.5 Software

The analysis was conducted in Python in a Jupyter Notebook. The code uses pandas and NumPy for data handling and numerical calculations; SciPy for statistical tests; statsmodels for MANOVA and related statistical procedures; scikit-learn for PCA; and Matplotlib and Seaborn for visualizations. The EFA extraction and rotation helpers were implemented in NumPy because the notebook notes compatibility problems with the factor_analyzer package in the execution environment.

5. Results

5.1 Exploratory Data Analysis

The combined dataset contains 6,497 observations and 11 physicochemical measurements, plus quality and color variables. No missing values were detected. There are 1,177 exact duplicate rows, representing 18.1% of observations; these were retained for the primary analysis and considered in sensitivity checks.

Variation differs considerably across variables. Based on the coefficient of variation, residual sugar varies the most (approximately 87%), followed by chlorides (63%), free sulfur dioxide (58%), total sulfur dioxide (49%), volatile acidity (48%), and citric acid (46%). Density varies least in relative terms (approximately 0.3%), followed by pH (approximately 5%). The raw variances differ by several orders of magnitude, supporting the decision to standardize variables before PCA and EFA.

The correlation structure shows some notable relationships but is not uniformly strong. There are 16 of 55 unique variable pairs with absolute correlation of at least .30 and three pairs with absolute correlation of at least .50. Density is highly predictable from the other measurements ($R^2 = .937$), reflecting strong overlap among density, alcohol, and residual sugar. Free and total sulfur dioxide are also correlated ($r \approx .72$).

Correlation patterns differ between red and white wines, suggesting that pooled structure partly reflects the difference between the two wine groups.

The data contain right-skewed variables and extreme observations. The EDA flagged 197 multivariate outliers across the 11-variable space using a chi-square threshold of 31.26, while the six-variable MANOVA diagnostics flagged 176 cases using a threshold of 22.46. These are separate analyses with different dimensions and should not be conflated. Several variables, including chlorides, fixed acidity, volatile acidity, and sulphates, have unusually high values. The observations were investigated through sensitivity analysis rather than automatically discarded.

EDA finding	Result	Interpretation
Sample composition	1,599 red; 4,898 white	Unequal group sizes; white wines are about three times as numerous
Missing data	0 detected	No missing-value imputation was required
Exact duplicates	1,177 (18.1%)	Retained in primary analysis; duplicate-free sensitivity check performed
Highest relative variability	Residual sugar CV ≈ 87%	Sugar levels vary substantially across samples
Density redundancy	R^2 = .937 from other variables	Density overlaps strongly with alcohol, sugar, and related measurements
Distribution shape	Strong right skew in several variables	Potential impact on covariance-based inference and factor estimation

5.2 MANOVA

The MANOVA found a statistically significant difference between the multivariate mean vectors of red and white wines across the six selected physicochemical variables. Wilks' Lambda was .237 and Pillai's Trace was .763, with $F(6, 6490) = 3{,}481.81$ and $p < .001$. The notebook reports a multivariate partial eta-squared of .763. Given the assumption violations and unequal group sizes, Pillai's Trace is emphasized as the primary statistic.

Variable	Red mean	White mean	F(1, 6495)	η^2	Cohen's d (Red − White)
Total sulfur dioxide	46.468	138.361	6252.80	.491	−2.278
Volatile acidity	0.528	0.278	4829.32	.426	2.002
Chlorides	0.087	0.046	2315.83	.263	1.386
Fixed acidity	8.320	6.855	2016.51	.237	1.293
Sulphates	0.658	0.490	2021.71	.237	1.295
Residual sugar	2.539	6.391	899.77	.122	−0.864

All six univariate ANOVAs were significant at $p < .001$ and remained significant under the Bonferroni threshold of approximately .0083; Welch tests agreed. Total sulfur dioxide and volatile acidity had the largest univariate effect sizes. White wines had a much higher mean total sulfur dioxide and residual sugar, whereas red wines had higher mean volatile acidity, chlorides, fixed acidity, and sulphates.

The discriminant-function results indicated that total sulfur dioxide and volatile acidity made the largest unique contributions to group separation. Residual sugar was significant in its univariate test but had little unique contribution in the combined discriminant score, suggesting that its separation overlaps with other variables.

Assumption checks found significant departures from multivariate normality, unequal covariance matrices (Box's M $\chi^2(21) = 7{,}222.4$, $p < .001$), and significant Levene tests for all six variables. Independence could not be verified because wine IDs and producer/batch identifiers were unavailable. Robustness checks preserved

the conclusion: Pillai's Trace ranged from .756 to .814, with $p < .001$ for the original, log-transformed, duplicate-free, and outlier-free analyses. The permutation test also supported separation ($p < .0005$, based on 2,000 permutations). These checks strengthen confidence in the existence of group differences, although assumption violations remain relevant when interpreting exact test statistics.

5.3 Principal Component Analysis

PCA was conducted on standardized versions of all 11 physicochemical variables. Bartlett's test of sphericity was significant, $\chi^2(55) = 37{,}094.6$, $p < .001$, indicating that the correlation matrix was not an identity matrix. However, the overall KMO was .405, below the usual .50 guideline. Density was a major reason: its R^2 when regressed on the other variables was .937, and KMO increased to .663 when density was removed. Because the assignment specified all 11 variables and PCA is a descriptive technique, all variables were retained, with the low KMO treated as a limitation.

Component	Eigenvalue	Variance explained	Cumulative variance
PC1	3.030	27.5%	27.5%
PC2	2.494	22.7%	50.2%
PC3	1.556	14.1%	64.4%
PC4	0.971	8.8%	73.2%

The Kaiser criterion, scree plot, and both parallel-analysis versions supported retaining three components. The first three components explain 64.4% of total standardized variance. The fourth component had an eigenvalue below one and did not exceed the parallel-analysis benchmark; retaining three components was therefore preferred over retaining four or five solely to reach a higher cumulative percentage.

Component	Major loadings (absolute value around .40 or above)	Interpretation
PC1 (27.5%)	Total sulfur dioxide .848; free sulfur dioxide .750; residual sugar .602; volatile acidity −.663; sulphates −.512; chlorides −.505; fixed acidity −.416	Sulfur dioxide/sugar versus acidity-and-salts profile; strongly associated with red–white separation
PC2 (22.7%)	Density .922; alcohol −.734; fixed acidity .531; residual sugar .521; chlorides .498	Density–sugar–alcohol balance
PC3 (14.1%)	Citric acid .737; fixed acidity .542; pH −.568	Acidity-related dimension

PC1 separates wine colors strongly: red wines had a mean score of −2.51 and white wines 0.82, with Cohen's $d = -3.39$ for red minus white. PC2 showed a smaller difference ($d = .68$), and PC3 showed little separation ($d = -.20$). Thus, the largest pooled component partly captures differences between wine colors as well as within-dataset chemical relationships.

The log-transformed PCA was highly similar to the original solution, with loading similarities of approximately .95–.99 for the first three components. Removing the red–white mean difference changed the structure: the density–sugar–alcohol dimension became the first component, and a fourth component with eigenvalue above one emerged. This suggests that the pooled PC1 is strongly influenced by the color contrast.

5.4 Exploratory Factor Analysis

EFA used Principal Axis Factoring with Promax rotation on the 11 physicochemical variables. Bartlett's test was significant, $\chi^2(55) = 37{,}094.6$, $p < .001$, but overall KMO was .405. Five variables had individual MSA values below .50: fixed acidity (.277), residual sugar (.294), density (.305), pH (.209), and alcohol (.273). These diagnostics indicate that the correlation matrix is only marginally suitable for common-factor modeling.

Three factors were retained based on the Kaiser criterion, standard parallel analysis, the scree plot, and interpretability. The factors account for approximately 53.1% of variance in the notebook's extraction summary. PAF was selected because the data were skewed and heavy-tailed; Promax was used because the factors were allowed to correlate.

Factor	Variance in extraction summary	Salient pattern loadings	Interpretive label
F1	23.34%	Total sulfur dioxide .79; free sulfur dioxide .65; residual sugar .46; fixed acidity −.77; chlorides −.48; sulphates −.46; volatile acidity −.43	Sulfur dioxide/sugar versus acidity-and-salts profile
F2	19.90%	Density 1.01; residual sugar .54; alcohol −.70	Density–sugar–alcohol balance
F3	9.83%	Citric acid .76; fixed acidity .59; pH −.52; volatile acidity −.41	Acidity-related factor

The factors were correlated: F1–F3 = .35, F2–F3 = .24, and F1–F2 = .05, supporting the use of oblique rotation. The factor structure resembles the PCA interpretation, but EFA focuses on common variance rather than decomposing all observed variance. Its reported 53.1% should therefore not be interpreted as directly equivalent to PCA's 64.4%.

Important diagnostic problems remain. Density produced a Heywood case: its communality was capped at .995 and its pattern loading on F2 was 1.01, which is not a valid correlation and signals an improper or unstable aspect of the factor solution. Sulphates and pH had low communalities (.22 and .29), and several variables cross-loaded. Log transformation produced high factor congruence (.97–.99), but removing density substantially changed F1. Removing the red–white difference also changed the first factor, indicating that F1 partly represents the pooled group contrast rather than a single stable latent chemical construct.

6. Integrated Discussion

The three methods offer complementary evidence about the wine dataset. MANOVA demonstrates that red and white wines differ substantially across six selected physicochemical measurements. PCA summarizes variation across all 11 measurements into three components explaining 64.4% of total variance. EFA identifies three exploratory common factors accounting for 53.1% in its extraction summary, while also revealing that the factor solution is less stable than the PCA solution because of low KMO, cross-loadings, and the density Heywood case.

The most consistent pattern across the methods concerns sulfur dioxide. Free and total sulfur dioxide contribute strongly to PC1 and F1, and total sulfur dioxide is the strongest univariate separator in MANOVA. White wines have a much higher average total sulfur dioxide concentration than red wines. This is compatible with sulfur dioxide's role in wine preservation, but the observational design cannot determine why the levels differ or whether color itself causes the difference.

A second recurring pattern is the density–residual sugar–alcohol relationship. Density and residual sugar load positively on PC2 and F2, while alcohol loads negatively. This reflects the interrelated physical and chemical properties of sugar, alcohol, and density. Because density is nearly predicted by the other variables, this dimension should not be over-interpreted as a fully independent latent construct.

A third pattern involves acidity. Citric acid and fixed acidity load positively on PC3 and F3, while pH loads negatively. MANOVA also finds higher average fixed and volatile acidity in red wines. These measurements

are chemically related but not interchangeable: pH describes hydrogen-ion activity, while acidity variables measure concentrations of specific acids or titratable components.

The methods are consistent in identifying broad patterns, but the interpretation must remain cautious. PCA components are mathematical combinations that maximize total variance; EFA factors are estimated constructs intended to explain shared covariance. Their labels are interpretive and do not prove the existence of distinct chemical mechanisms. The pooled PCA and EFA also partly capture the red–white contrast, so a factor or component that appears prominent in the pooled sample may not have the same structure within each wine color.

7. Conclusion

The analysis found strong and statistically significant physicochemical differences between red and white Portuguese Vinho Verde wines. MANOVA reported Pillai's Trace = .763, $F(6, 6490) = 3{,}481.81$, $p < .001$, and all six follow-up tests remained significant after Bonferroni correction. White wines had higher mean total sulfur dioxide and residual sugar, while red wines had higher mean volatile acidity, chlorides, fixed acidity, and sulphates.

PCA retained three components explaining 64.4% of total variance: a sulfur dioxide/sugar versus acidity-and-salts contrast, a density–sugar–alcohol balance, and an acidity-related dimension. EFA retained three exploratory factors explaining approximately 53.1% according to the extraction summary, with similar broad interpretations but important warnings from the low KMO (.405), low communalities for some variables, cross-loadings, and a density Heywood case.

Overall, sulfur dioxide, acidity-related measurements, and the density–sugar–alcohol relationship are central to the multivariate structure of this dataset. These findings are descriptive rather than causal, and the chemical measurements do not by themselves explain sensory quality. Future work could examine color-specific structures and explicitly model the quality score while accounting for the limitations of the observational data.

8. References

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Note: Confirm the required citation style with your instructor. The software and package descriptions in the Methodology section are based on the libraries used in the submitted notebook.